

	CAPM (Market Factor)	Multifactor Models
Lesson 4	The market factor is priced in equilibrium under the CAPM assumptions.	Risk premiums exist for each factor assuming no arbitrage or equilibrium.
Lesson 5	Risk of an asset is measured by the CAPM beta.	Risk of an asset is measured in terms of the factor exposures (factor betas) of that asset.
Lesson 6	Assets paying off in bad times when the market return is low are attractive, and these assets have low risk premiums.	Assets paying off in bad times are attractive, and these assets have low risk premiums.

The \$64,000 question with multifactor pricing kernel models is: how do you define bad times? For the average investor who holds the market portfolio, the answer is when an extra \$1 becomes very valuable. This interprets the SDF as the marginal utility of a *representative agent*. We will come back to this formulation in chapter 8 when we characterize the equity risk premium. Times of high marginal utility are, for example, periods when you’ve just lost your job so your income is low and any extra dollars are precious to you. Your consumption is also low during these times. In terms of the average, representative consumer, this also corresponds to a macro factor definition of a bad time: bad times are when GDP growth is low, consumption is low, or economic growth in general is low. Times of high marginal utility could also be defined in relative terms: it could be when your consumption is low relative to your neighbor or when your consumption is low relative to your past consumption. In chapter 2, we captured the former using a catching up with the Joneses utility function and the latter with a habit utility function.

During 2008–2009, the financial crisis was a bad time with high volatility and large financial shocks. So volatility is an important factor, and the next chapter shows that many risky assets perform badly when volatility is high. Factors can also be tradeable, investment styles. Some of these include liquid, public market asset classes like bonds and listed equities. Others include investment styles that are easily replicable and that can be implemented cheaply (but often are not when they are delivered to customers) and in scale, like value/growth strategies.⁹

⁹There is a third type of factor based solely on statistical principal components, or similar (dynamic) statistical factor estimations of the APT. A pioneering example of these is Connor and Korajczyk (1986). These generally lack economic content, and so I do not discuss them here.